

Muhammad Azzam Hanif Prasetyo

Sendai, Japan 981-0935 • orcid.org/0009-0006-1665-3526 • hanif.prasetyo@gmail.com • linkedin.com/in/hanifprs

SUMMARY

Researcher with focus on ocean remote sensing, ecology, climate and physical oceanography for the past $\pm$ 5 years—experienced working at various international research centres and projects in Indonesia, South Korea and Japan. Currently at the Centre for Atmospheric and Oceanic Research (CAOS), satellite oceanography laboratory as a master's student at Tohoku University.

Key qualification

Remote sensing:

- Supported researchers from KIOST and JAMSTEC in the calibration and validation (Cal/Val) of GOCI-II and GCOM-C ocean-colour satellites in case-1 and case-2 waters in Indonesian waters.
- Experienced in analysing changes in benthic habitat on interseasonal and interannual timescales using hyperspectral sensors and satellite imagery. Additional experience on LULC and carbon stock analysis to support environmental monitoring and conservation objectives in various places in Indonesia using multispectral and lidar drones.

Marine ecology:

- Certified in scientific diving from Scuba Schools International (SSI) with +50 dive logs.
- Experienced in data collection, management, and analysis on coral reefs, seagrass, and mangroves for ecology monitoring.

Climate and physical oceanography:

- Experienced in detecting marine heatwave phenomena around the maritime continent and analysing their impact on coral reef coverage on interseasonal and interannual timescales.

EDUCATION

TOHOKU UNIVERSITY

Geophysics, GPA: 3.3 / 4.00

Sendai, Japan

October 2025-Present

- Research focus: The relationship between coral bleaching and marine heatwaves measured with hyperspectral sensors.
- MEXT scholarship awardee

DIPONEGORO UNIVERSITY

Marine Science, GPA: 3.72 / 4.00

Semarang, Indonesia

August 2019-January 2024

- Undergraduate Thesis on “Comparison of Coral Cover Pre and Post-2016 Mass Coral Bleaching in Nusa Dua, Bali”

PUBLICATION

The Possible Contribution of Marine Heatwave on The Declining Coral Reef Cover in Nusa Dua, Bali (Published in International Journal of Remote Sensing) DOI: doi.org/10.1080/01431161.2025.2592906 November 2025

Genetic Identification of Symbiotic Algae in Hard Corals from Maratua Island (ongoing manuscript and will be submitted to Biodiversitas Journal of Biological Diversity) April 2024

WORK EXPERIENCE

KOREA-INDONESIA MARINE TECHNOLOGY COOPERATION RESEARCH CENTER (MTCRC)

Jakarta, Indonesia

Remote Sensing Surveyor:

August 2024 - August 2025

Main responsibility:

- Collect, process and report field observation data in cooperation with Korea-Indonesia MTCRC established and on-going ODA project, as well as provide support for additional activities with government institutions and universities in Indonesia.
- Maintain and operate instruments related to the remote sensing and physical oceanography division. For example: drone, CTD, sub-bottom profiler, multi-beam/single-beam echosounder, RTK GNSS, lidar, multispectral and hyperspectral sensors.
- Supporting stakeholders as a speaker or instructor in training and workshops related to remote sensing and physical oceanography surveys.

Project experience:

- Performed and processed in-situ remote sensing reflectance (Rrs) measurement using TriOS RAMSES radiometers for ocean colour satellite calibration and validation (Cal/Val) on GOCI-II (South Korea) and GCOM-C (Japan) satellite in Indonesia, Case 1 and Case 2 waters. In collaboration with KIOST, JAMSTEC, JAXA and IPB University.
- Performed benthic habitat mapping in shallow water using in-situ USSIMO submersible hyperspectral radiometers correspond with the deployment of dual-red edge MicaSense multispectral drone and Sentinel-2 satellite for analysing

temporal changes. In charge as the diver deploying the sensor underwater and has experience in drone deployment until data processing from raw radiance/irradiance to calculated remote sensing reflectance (Rrs).

- Performed deployment and processing for Lidar mapping using CHCNAV AA450 mount on DJI Matrice 300 for analysing disaster management in coastal regions vulnerable to tsunamis.
- Performed carbon stock analysis estimation on Indonesia's newly established district capital in Kalimantan Island in cooperation with Gadjah Mada University. The project concludes with obtaining below-ground sediment samples with coring techniques and laboratory process using the Total CN Analyzer before creating above and below-ground carbon stock estimation.

RESEARCH PROJECT EXPERIENCE

DIPONEGORO UNIVERSITY

Semarang, Indonesia

Research Assistant for e-Asia Joint Research Project on Marine Heatwaves in the Western Pacific: October 2023 - August 2024

- Perform benthic habitat mapping using Sentinel-2 imagery with a random forest classification scheme to detect changes in coral coverage at interseasonal and interannual timescales, corresponding with the intensity and occurrence of marine heatwaves in the area.
- Conducted ecological assessment on coral reef ecosystems, which included collecting data regarding coral coverage, inherent optical properties (Chl and CDOM), total suspended solids (TSS), and water quality parameters (pH, DO, salinity, and SST). Coral coverage data is analysed using CPCE to create a ground truth reference for the classification model and ecological thresholds for the Marine Protection Area report.
- Serve as the first author in paper related to the topic (listed in the publication section), published in the International Journal of Remote Sensing as part of the e-Asia Joint Research Project with universities from Japan, USA, Philippines and Thailand.

Research assistant for Marine Sciences Accelerating Program (MSAP) research grant:

August 2021 - February 2022

- Conducted a genetic analysis procedure including DNA Extraction using an animal extraction kit, PCR with electrophoresis methods, and post-sequence DNA analysis using MEGA X software.
- Wrote proposal and report findings as part of the weekly report for the MSAP research grant project.
- Collecting field samples of hard coral in Maratua Island to assess its genetic biodiversity.

Teaching assistant for Marine Sciences Department in various subjects:

January 2021 - August 2023

- Provided training for students on coral reef survey methods, data management, and analysis with interactive assignments.
- Created a teaching module on topography and bathymetry maps using in-situ data that improved students' understanding and was later adopted by the next batch of assistants.
- Provided training for students on coral reef survey methods, data management, and analysis with interactive assignments.

CERTIFICATION

Graduate Record Examination (GRE) - ETS	November 2024
International English Language Testing System (IELTS) Band 7 - IDP Education	April 2024
Science of Diving Specialist - SSI Scuba Schools International	October 2021
One Star Open Water Scuba Diver - CMAS	August 2020
Land Navigation Speciality - Wanadri	March 2017

SKILLS AND INTEREST

Interest	: Radiative transfer in shallow water, hyperspectral remote sensing for benthic habitat mapping and ocean colour on global coupled BGC models
Technical skills	: Ocean colour instrument measurement, drone pilot, GNSS measurement, laboratory analysis and hyperspectral sensor deployment
Non-technical skills	: Leadership, Analytical Thinking, Agile Methodologies, Scrum Process Management
Software	: ENVI, Arc GIS, QGIS, SNAP, Google Earth Engine, WASI, Pix4D, Sea Das, ODV, Mike, and Delft 3D
Programming	: Python and JavaScript

REFERENCE

Prof. Sayaka Yasunaka | Research Supervisor at Tohoku University | yasunaka@tohoku.ac.jp

Prof. Anindya Wirasatriya | Supervisor and principal investigator on e-Asia Joint Research | anindyawirasatriya@lecturer.undip.ac.id

Dr. Hansan Park | Co-Director of Korea-Indonesia MTCRC | hansanpark@kiost.ac.kr